

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

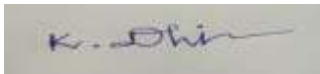
Question Count: 5

Code of Conduct

I K.Dhilip/192365013 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

A hypervisor is virtualization software that allows multiple virtual machines (VMs) to run on a single physical server. Hypervisors are mainly classified into Type-1 (Bare-Metal) and Type-2 (Hosted) hypervisors.

Type-1 Hypervisor

A Type-1 hypervisor is installed directly on the physical hardware without requiring a host operating system. Since it communicates directly with the hardware, it provides excellent performance and efficient resource utilization. It is commonly used in enterprise cloud environments because it supports advanced features such as live migration, high availability, and fault tolerance.

Type-2 Hypervisor

A Type-2 hypervisor runs on top of an existing operating system. The host operating system acts as an intermediary between the hardware and the virtual machines. Although it is easy to install and suitable for development or testing purposes, it has lower performance due to additional overhead.

Comparison

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Directly on hardware	On top of host OS
Performance	Very High	Moderate
Security	More Secure	Less Secure
Resource Usage	Efficient	Higher Overhead
Scalability	Excellent	Limited
Typical Use	Data centers and cloud	Personal Computer and testing

Performance Analysis

Type-1 hypervisors provide better CPU, memory, and storage performance because they access hardware directly. In contrast, Type-2 hypervisors rely on the host operating system, resulting in additional processing delays.

Security Analysis

Type-1 hypervisors have a smaller attack surface since they do not depend on a host operating system. This reduces security vulnerabilities. Type-2 hypervisors inherit security risks from the host operating system, making them more vulnerable to malware and attacks.

Manageability Analysis

Type-1 hypervisors include enterprise management features such as centralized administration, automatic load balancing, live migration, snapshots, and disaster recovery. Type-2 hypervisors provide only basic management features and are not suitable for large-scale cloud deployments.

Justification

Mission-critical applications require high availability, fast performance, strong security, and reliable management. Type-1 hypervisors satisfy all these requirements and are widely used by cloud providers because they minimize downtime and maximize resource utilization.

Conclusion

For hosting mission-critical cloud applications, Type-1 hypervisors are the better choice because they provide higher performance, stronger security, better scalability, and advanced management capabilities.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Cloud providers often run multiple customer workloads on the same physical server to reduce infrastructure costs. However, these workloads must remain completely isolated to ensure security and privacy. Virtualization architecture provides this isolation through the use of virtual machines and the hypervisor.

Each department, such as Finance, Healthcare, and Student Portal, is assigned its own virtual machine. Every virtual machine has its own operating system, memory, CPU allocation, storage, and network configuration. Even though all virtual machines share the same hardware, they operate independently.

The hypervisor manages hardware resources and prevents one virtual machine from directly accessing another. If the Finance VM experiences high CPU usage or crashes, the Healthcare and Student Portal VMs continue to operate normally.

Virtual networking also improves isolation. Technologies such as VLANs and virtual switches ensure that network traffic from one department cannot be accessed by another. Similarly, storage virtualization provides separate virtual disks so that data belonging to different departments remains isolated.

Benefits of Virtualization Isolation

- Better security between workloads.
- Efficient utilization of hardware resources.
- Easier management and maintenance.
- Independent software updates.
- Improved scalability and flexibility.

Major Attack Surfaces

1. Hypervisor (VM Escape Attack)

In a VM escape attack, an attacker exploits a vulnerability in the hypervisor to gain access to the host machine or other virtual machines. Since the hypervisor controls all VMs, compromising it can affect the entire system.

2. Virtual Network Attack

Improperly configured virtual switches or VLANs may allow attackers to intercept network traffic or perform unauthorized access between virtual machines.

Security Measures

- Keep the hypervisor updated with security patches.
- Use strong authentication and access control.
- Enable encryption for stored and transmitted data.
- Apply network segmentation using VLANs.
- Monitor virtual machines continuously.

Conclusion

Virtualization architecture provides strong workload isolation by separating CPU, memory, storage, and network resources. Proper configuration and security practices help protect against hypervisor and virtual network attacks.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

The given utilization values indicate that the virtualized host is under heavy load.

Analysis of CPU Utilization (92%)

The CPU utilization is extremely high. This means the processor is busy most of the time, leaving very little capacity for additional workloads. As a result, applications may become slow and users may experience delays.

Analysis of Memory Utilization (88%)

Memory utilization is also very high. When physical memory becomes insufficient, the system may use virtual memory or swapping, which significantly reduces performance.

Analysis of Storage I/O Latency (35 ms)

A storage latency of 35 milliseconds is much higher than the recommended value for cloud environments. High latency indicates slow read and write operations, affecting database performance and virtual machine startup.

Analysis of VM Boot Time (170 Seconds)

A boot time of nearly three minutes suggests storage and CPU bottlenecks. The system requires more time to allocate resources and load operating systems.

Likely Bottlenecks

i.) CPU (92%)

- CPU is nearly saturated.
- Causes slow application response.

ii.) Memory (88%)

- Very high utilization.
- May cause swapping and reduced performance.

iii.) Storage I/O Latency (35 ms)

- Storage is slow.
- Delays VM startup and disk operations.

iv.) VM Boot Time (170 s)

- Indicates storage and CPU bottlenecks.

Corrective Actions

1. Increase CPU cores or migrate some virtual machines to another host.
2. Upgrade the server memory.
3. Replace HDD storage with SSD or NVMe drives.
4. Enable load balancing and dynamic resource scheduling.
5. Remove idle or unnecessary virtual machines.
6. Monitor system performance continuously.

Expected Improvements

- Reduced CPU utilization.
- Faster VM startup.
- Lower storage latency.
- Better application response time.
- Improved user experience.

Conclusion

The major bottlenecks are high CPU usage, memory pressure, and slow storage performance. Upgrading hardware resources and balancing workloads will significantly improve overall system performance.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

A Software Defined Data Center (SDDC) is a modern data center in which compute, storage, and networking resources are fully virtualized and managed through software. Unlike traditional data centers that require manual hardware configuration, SDDCs provide automation, flexibility, and faster service delivery.

Compute Virtualization

Compute virtualization allows multiple virtual machines to run on a single physical server. New virtual machines can be created within minutes instead of purchasing and installing new hardware. Resources can also be automatically increased or decreased according to workload demand.

Storage Virtualization

Storage virtualization combines multiple storage devices into a single logical storage pool. Administrators can allocate storage dynamically without changing physical hardware. It also simplifies backup, disaster recovery, and storage expansion.

Network Virtualization

Network virtualization creates logical networks independent of physical infrastructure. Virtual switches, routers, firewalls, and load balancers can be deployed through software. This enables faster network configuration and improved security through network segmentation.

Advantages over Traditional Data Centers

Traditional Data Center	Software Defined Data Center
Manual configuration	Automated management
Hardware dependent	Software controlled
Slow deployment	Rapid deployment
Limited scalability	Highly scalable
Higher operational cost	Lower operational cost

Benefits of SDDC

- Faster resource provisioning.
- Improved resource utilization.
- Reduced operational costs.
- Better disaster recovery.
- Automatic scaling based on demand.
- Simplified infrastructure management.

Conclusion

Software Defined Data centers improve agility by virtualizing compute, storage, and networking resources. Automation and centralized management allow organizations to deploy applications faster, reduce costs, and respond quickly to changing business requirements.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Many organizations use services from multiple cloud providers to improve flexibility and avoid dependence on a single vendor. However, different providers often use different identity management systems. This leads to identity mismatches, making authentication and authorization difficult.

Problem Analysis

Each cloud provider may maintain its own user accounts, authentication methods, and access policies. As a result:

- Users must remember multiple usernames and passwords.
- Identity information becomes inconsistent.
- Access permissions may not be synchronized.
- Security risks increase due to duplicate accounts.

These issues reduce user convenience and make administration more complex.

Identity Federation

Identity federation establishes trust between different cloud providers. A user authenticates once with a trusted Identity Provider (IdP), and the authentication is accepted by all participating cloud services.

Secure Identity Design

A secure multi-cloud identity solution should include:

- **Single Sign-On (SSO):** Users log in once and access multiple cloud services without repeated authentication.
- **Federated Identity:** A common digital identity is shared across providers.
- **Standard Authentication Protocols:** Use SAML, OAuth 2.0, or OpenID Connect for secure identity exchange.
- **Role-Based Access Control (RBAC):** Permissions are assigned based on user roles instead of individual accounts.
- **Multi-Factor Authentication (MFA):** Adds an extra layer of security beyond passwords.
- **Encryption:** Protect identity information while it is stored and transmitted.
- **Audit Logs:** Record authentication and access activities for monitoring and compliance.

Privacy Protection

- Share only the minimum user information required.
- Encrypt sensitive personal data.
- Follow the principle of least privilege.
- Regularly review access permissions.

Benefits

- Simplified user authentication.
- Stronger security.
- Better privacy protection.
- Easier administration.
- Improved compliance with organizational policies.

Conclusion

Identity mismatches are a common challenge in multi-cloud environments. Implementing identity federation with SSO, standardized authentication protocols, RBAC, MFA, encryption, and audit logging provides secure, seamless, and privacy-preserving access across multiple cloud providers. This approach improves both security and user experience while reducing administrative complexity.

